

that the attitudes of low knowledge subjects were affected by argument length, but the attitudes of high knowledge subjects were not (see bottom panel of Figure 6-9). In sum, low knowledge subjects' attitudes were affected by the simple cue of message length, but high knowledge subjects' used their prior knowledge in an attempt to defend their attitudes. They were more successful in doing this when the arguments in the message were weak rather than strong.⁴

Retrospective

In this chapter we explored the tradeoff between argument processing (whether relatively objective or biased) and the operation of peripheral cues. We have seen that when a person's ability and/or motivation to process issue-relevant arguments is attenuated, peripheral cues in the persuasion context become more important determinants of influence. Affective, cognitive, and behavioral cues have all proven effective when the elaboration likelihood was low. Furthermore, these cues have resided in the source (e.g., credibility, likability), the message itself (e.g., mere number of arguments), the recipient (feedback of bodily responses, mood), and the persuasion context (e.g., pleasant music, audience reactions). As the elaboration likelihood is increased, however, peripheral cues become less important. In short, in this chapter we have seen how a wide variety of individual (e.g., prior knowledge, need for cognition) and situational (e.g., distraction, message modality) variables moderate the route to persuasion by affecting the likelihood of issue-relevant elaboration and the reliance on peripheral cues. In the next chapter we examine the differential consequences of these two routes to persuasion.

⁴For ease of exposition we have graphed the data based on a median split on prior knowledge (Wood, personal communication, October 18, 1984) rather than the three-way split reported in the published article. As might be expected, the three-way split only enhances the differences between high and low knowledge groups, though the median split is based on a larger sample per cell.

Chapter 7

Consequences of the Route to Persuasion

Introduction

In the preceding chapters of this monograph we have outlined how the Elaboration Likelihood Model accounts for the initial attitude changes induced by persuasive messages, and we have reviewed the evidence for the ELM. We have seen that there are two qualitatively different routes to persuasion, which are moderated by a continuum of elaboration likelihood. When the elaboration likelihood is high, people follow the central route to persuasion as they attempt to scrutinize the issue-relevant information presented. This processing may proceed in either a relatively objective or a relatively biased manner. When the elaboration likelihood is low, people follow the peripheral route to persuasion. Under this second route, attitudes are influenced by relatively simple cues in the persuasion context that either become directly associated with the advocacy or allow an inference as to the likely correctness or desirability of a particular attitude position without necessitating a personal evaluation of the issue-relevant arguments presented. The last postulate of the ELM proposes that there are at least three important consequences of the route to persuasion: attitudes formed or changed via the central route will show greater temporal persistence, resistance to counterpropaganda, and prediction of behavior than attitudes formed or changed via the peripheral route. In Chapter 1 we provided the justification for this postulate. In this chapter we review the evidence for this proposition.

Persistence of Persuasion

Enhanced Thinking Produces Persistence

If extended issue-relevant thinking increases the temporal persistence of opinion change, then conditions that foster issue-relevant elaboration

should be accompanied by greater attitudinal persistence than conditions that minimize elaboration. In this section we review some evidence bearing on this notion.

Research on Role Playing

Among all the ways to change attitudes, one of the influence paradigms requiring the most issue-relevant thinking to produce persuasion is "role playing." In role playing research people are required to generate or improvise their own arguments for a message (e.g., King & Janis, 1956). Importantly, research indicates that to the extent that people have sufficient knowledge and skill to generate their own messages, the attitude changes induced by these messages are especially persistent.

In an illustrative study, Watts (1967) had subjects read a communication or actively generate a passage on an assigned topic. The persuasive messages employed in the experiment were developed in pretesting so that they would produce the same amount of initial attitude change as would the self-generation of a communication. The pretesting was apparently successful in that subjects who passively read the communication showed an amount of initial attitude change equivalent to that of subjects who generated their own messages. Both groups showed significantly more attitude change than controls. On a delayed measure of attitudes, taken 6 weeks later, however, subjects in the self-generation conditions showed total persistence of attitude change, whereas subjects in the passive reading conditions showed total decay.

Elms (1966) conducted a conceptually similar study. After reading a brief anti-smoking pamphlet, some subjects were asked to play the role of a person who had a serious disease and quit smoking. These subjects were told that it was their job to generate a message that would convince their best friend to quit. These subjects were also told to write down all of the reasons they could think of for giving up smoking. Another group of subjects, playing the role of the best friend, passively listened to the message generated by the active subjects after they too had read the anti-smoking pamphlet. Finally, the passive subjects also wrote down all of the reasons they could think of to stop smoking. On the immediate test of opinion, both active and passive subjects showed equivalent amounts of attitude change in the anti-smoking direction. On a measure taken 3 weeks later, however, the active subjects were significantly more anti-smoking than the passive group. Since the role playing subjects engaged in a more active and effortful issue-relevant elaboration than passive subjects, the ELM would expect their attitude changes to show greater temporal persistence. One factor that is a likely contributor to this greater persistence is that self-generated arguments are more accessible than arguments generated by others (Greenwald & Albert, 1968; Slameka & Graf, 1978).

Research on Anticipatory Shifts

Although in the role playing studies, subjects are instructed to think about an issue position, in other research the experimental conditions elicit issue-

relevant thought spontaneously. For example, in research on anticipatory attitude shifts, subjects are induced to expect to discuss an issue or receive a message on some topic, and attitudes are measured prior to the discussion or message presentation (Cialdini & Petty, 1981). In one anticipatory change study relevant to persistence (Cialdini, Levy, Herman, Kozlowski, & Petty, 1976), college students were led to believe that they would discuss a campus issue with another student who held a position opposite to their own. Subjects were told that the discussion would take place either immediately or one week later, and the issue to be discussed was either one that was personally important to the students or unimportant. While waiting for the discussion to begin, subjects listed their thoughts on the issue and then reported their attitudes. Although subjects in all conditions showed some anticipatory shifting of their positions, only one group of subjects maintained their new issue positions after they had been informed that the discussion was cancelled. This group, subjects who expected to immediately discuss a personally important issue, was presumably the most motivated to undertake the cognitive work necessary to prepare for the discussion. Consistent with this analysis, these subjects listed significantly more thoughts that supported their own positions in anticipation of the discussion than subjects in the other cells.

Testing the ELM Analysis of Persistence

In both the role playing research and the research on anticipatory shifts, attitude changes that were accompanied by considerable issue-relevant cognitive activity led to more persisting shifts than changes induced with less issue-relevant thinking. However, in both of these paradigms, no persuasive messages were presented. According to the ELM, the same result should hold if the attitude changes resulted from exposure to a persuasive communication. Specifically, the greater the elaboration of the message arguments, the more persistent the resulting attitude change should be. The accumulated persuasion literature is quite consistent with the idea that conditions that foster people's ability or motivation to engage in issue-relevant cognitive activity also enhance the persistence of persuasion. Thus, research has shown that using more interesting or involving issues about which subjects have more knowledge (e.g., Ronis, Baumgardner, Leippe, Cacioppo, & Greenwald, 1977), providing more time to think about the message (Mitnick & McGinnies, 1958), leading people to believe that they will be interviewed on the attitude issue (Chaiken, 1980), increasing message repetition (Johnson & Watkins, 1971), and reducing distraction (Watts & Holt, 1979), all are associated with increased temporal persistence of attitude change (see Cook & Flay, 1978; Petty, 1977; for reviews). To provide a more direct test of the ELM view of persistence, however, we attempted to induce comparable attitude changes under the central and the peripheral routes, and then track attitude persistence over time (Petty, Cacioppo, Haugtvedt, & Heesacker, 1986; Experiment 1).

Method. In our study on attitude persistence, 100 students participated in a 2 (Personal relevance: high or low) $\times$ 2 (Message type: strong arguments-positive cue or weak arguments-negative cue) $\times$ 2 (Time of assessment: immediate or delayed) mixed model factorial design. All of the subjects in the experimental conditions listened to three persuasive messages. Each message began with a description of the origin of the message and provided a brief biography of the message source. The first two messages served as filler material and were identical for all subjects. The third message contained the experimental manipulations. Half of the subjects were led to believe that the advocacy concerned an imminent change in policy (the institution of senior comprehensive exams) at their own university (high relevance), and half were led to believe that the advocacy concerned the same proposed policy change at a distant university (low relevance). Half of the students received a message from a very prestigious and credible source, which provided six strong arguments in support of instituting senior comprehensive exams, and half of the students received a message from a low prestige, inexperienced source, which provided six weak arguments in support of the exams. An additional group of 25 subjects received an irrelevant third message and served as an attitude-only control. Subjects' attitudes were first assessed immediately after the message presentation. Then, from 10 to 14 days following message exposure, subjects were called and asked their opinions concerning a number of campus issues, including the general idea of senior comprehensive exams.

Based on our previous research on personal relevance (Petty & Cacioppo, 1979b; Petty, Cacioppo, & Goldman, 1981; see Chapters 4 and 6) and additional pilot testing of the sources and messages under high and low relevance conditions, we expected that both the high and low involvement groups who received the positive source-strong arguments message would show equivalent amounts of initial persuasion. However, the change in the high relevance group should be based mostly on a careful evaluation and elaboration of the strong issue-relevant arguments (central route), whereas the change in the low relevance group should be based mostly on the positive source cue (peripheral route). Similarly, the rejection of the advocacy in the high and low relevance groups exposed to the negative source-weak arguments message should be equivalent initially, but in the high involvement group the rejection should be based mostly on scrutiny of the weak arguments, whereas in the low relevance group the rejection should be based mostly on the negative source. Our major prediction was that attitudes changed via the central route (high relevance conditions) would show greater temporal persistence than attitudes changed to the same degree, but via the peripheral route (low relevance conditions).

Results. An analysis of subjects' immediate and delayed attitudes provided support for the ELM predictions regarding persistence. On the immediate measure of attitudes, both high and low relevance groups of

subjects exposed to the positive message/source were more favorable than controls, and both groups of subjects exposed to the negative message/source were less favorable than controls. More interestingly, however, the degree of personal relevance had an impact on whether or not these initial attitudes persisted. Specifically, an analysis of variance on the initial and delayed attitudes of high relevance subjects revealed only a main effect for type of communication. The positive source-strong arguments message was more effective than the negative source-weak arguments message both initially and at the delayed testing. An analysis on the attitudes of low relevance subjects, however, revealed a Message type $\times$ Time of measurement interaction (see Figure 7-1). For these subjects, the initial difference between the two message conditions was no longer apparent at the delayed testing. Specifically, low involvement subjects exposed to the positive

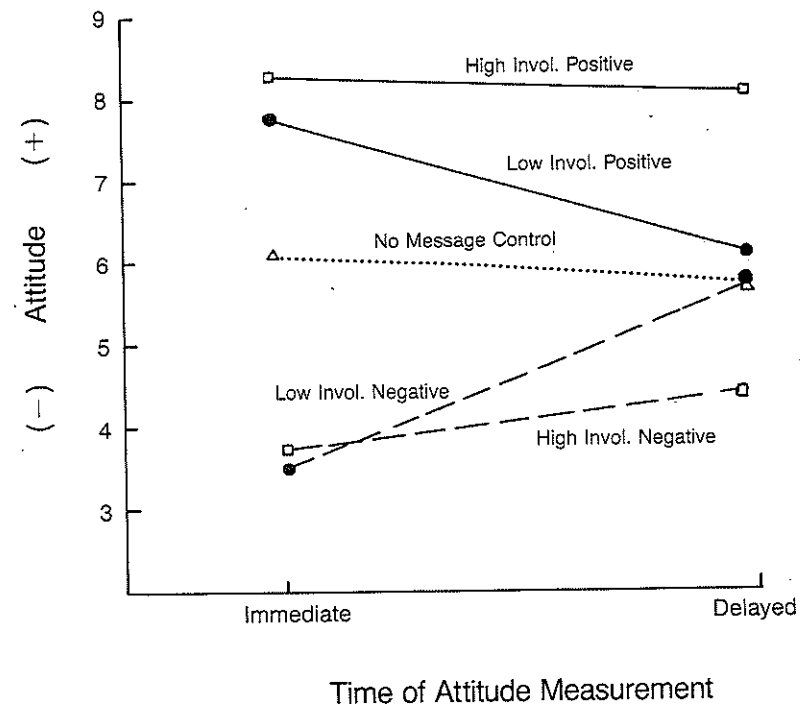


Figure 7-1. Immediate and delayed postmessage attitudes as a function of personal relevance (involvement) and type of communication. The figure shows that attitude changes, whether induced by a positive (strong arguments, high credible source) or a negative (weak arguments, low credible source) communication, persist longer when initially processed under high rather than low involvement (elaboration) conditions (data from Petty, Cacioppo, Haugtvedt, & Heesacker, 1986, Experiment 1).

message/source were less favorable at the delayed than the initial assessment, but those exposed to the negative message/source were more favorable at the delayed than the initial assessment. In short, those subjects who formed their initial attitudes based on a careful consideration of issue-relevant arguments showed greater persistence of attitude change (whether favorable or boomerang) than those subjects whose initial attitudes were based primarily on the source cue.

Alternative Models of Attitude Persistence

Message Learning Model

Before proceeding with other consequences of the route to persuasion, it would be useful to compare the ELM to other models of the persistence of attitude change. Perhaps the most widely studied model is the "Yale" message learning approach as pioneered by Carl Hovland and his colleagues (e.g., Hovland, Lumsdaine, & Sheffield, 1949; Hovland, Janis, & Kelley, 1953) and elaborated by William McGuire and others (e.g., Watts & McGuire, 1964; see also, Miller & Campbell, 1959; Eagly, 1974; Eagly & Himmelbarb, 1974). As McGuire (1968) indicates, use of the message learning approach involves "predicting how any independent variable in the communication situation (such as order of presentation, source credibility, . . . etc.) will be related to attitude change by analyzing that variable's likely impact on learning the message contents. Hence, if there is a primacy effect in learning, then one is predicted also in retention of message content; if there is a negatively decelerated decay in retention of message content, a like function is predicted as regards the persistence of induced attitude change." (p. 179).

However, reviewers of the literature on retention of message content and persistence have concluded that there appears to be little link between retention of specific message arguments and persistence of persuasion (e.g., Brock & Shavitt, 1983; Fisk, 1981; Greenwald, 1968; McGuire, 1969; Miller & Colman, 1981). As Cook and Flay (1978) summarized: "It seems unlikely that retention of message details and delayed attitude are related in any simple way" (p. 23). Studies examining correlations between message retention and delayed attitude have found the correlations to be very modest, nonsignificant, or even negative (e.g., Cacioppo & Petty, 1979b; Insko, 1964; Watts & McGuire, 1964). Furthermore, studies examining the decay function of attitudes and message retention have revealed that the two measures may not follow the same temporal pattern (e.g., Papageorgis, 1963). In his comprehensive review of the attitudes literature in 1969, McGuire concluded that the "learning theory approach to attitude change seems to us to be acquiring more the status of a fertile error" (p. 266).

Cognitive Response Model

In an attempt to salvage a learning model of persistence, Greenwald (1968) proposed a "cognitive response" model which argued that just as initial

persuasion depended upon the cognitive responses (idiosyncratic associations) to the message content, persistence of persuasion depended upon the learning and retention of these cognitive responses. In a test of this idea, Love and Greenwald (1978) had undergraduates complete an opinion pretest, read one of two messages, list their cognitive reactions to each argument in the message, complete an opinion posttest, and then attempt to recall the arguments in the message and their reactions. One week later, subjects again completed an opinion posttest and attempted to recall the message arguments and their cognitive responses. Analyses employing partial correlations (initial pretest was partialled out) revealed that although the delayed opinion measure could be predicted on the basis of the cognitive responses recalled at the delayed testing session (number of favorable minus unfavorable reactions recalled), the measure of number of message arguments recalled was not significantly related to delayed opinion. Although this study provides evidence consistent with Greenwald's cognitive response hypothesis, the evidence is correlational and therefore questions about causal direction remain (Cook & Flay, 1978).

In a study designed to provide an experimental test of the cognitive response hypothesis, Petty (1977) exposed undergraduates to one of two messages on the topic of increasing the legal driving age to 21. One message presented 5 highly persuasive (strong) arguments and the other contained 5 less persuasive (weak) arguments in favor of the proposal. A third (control) group of subjects read 5 neutral statements about driving. Each of the arguments and statements was only one sentence long (14 to 28 words). After reading the appropriate message, subjects were asked to list five of their own thoughts about the proposal to raise the driving age. The major manipulation in this study was whether subjects were asked to memorize the specific arguments in the message or their own thoughts on the issue with all subjects being given as much time as was needed to memorize the material close to verbatim. One measure of attitudes was taken right after the initial memorization task, and a second measure was taken one week later. At the delayed session, subjects were also requested to recall the arguments/statements and the thoughts generated the previous week.

The results of this study revealed that subjects receiving the strong arguments generated more favorable thoughts and fewer unfavorable thoughts than subjects receiving the weak arguments or neutral statements. In addition, subjects receiving the strong arguments were more favorable toward raising the driving age than the remaining subjects on the initial measure of opinion (see top panel of Figure 7-2). More interestingly, when attitudes were assessed one week later, only those subjects who were exposed to the strong arguments and who memorized their cognitive responses persisted in their new opinions. Subjects who were exposed to the strong arguments but who memorized the exact message statements rather than their own reactions to the message were no longer different from the controls or subjects who received the weak arguments (see bottom panel of

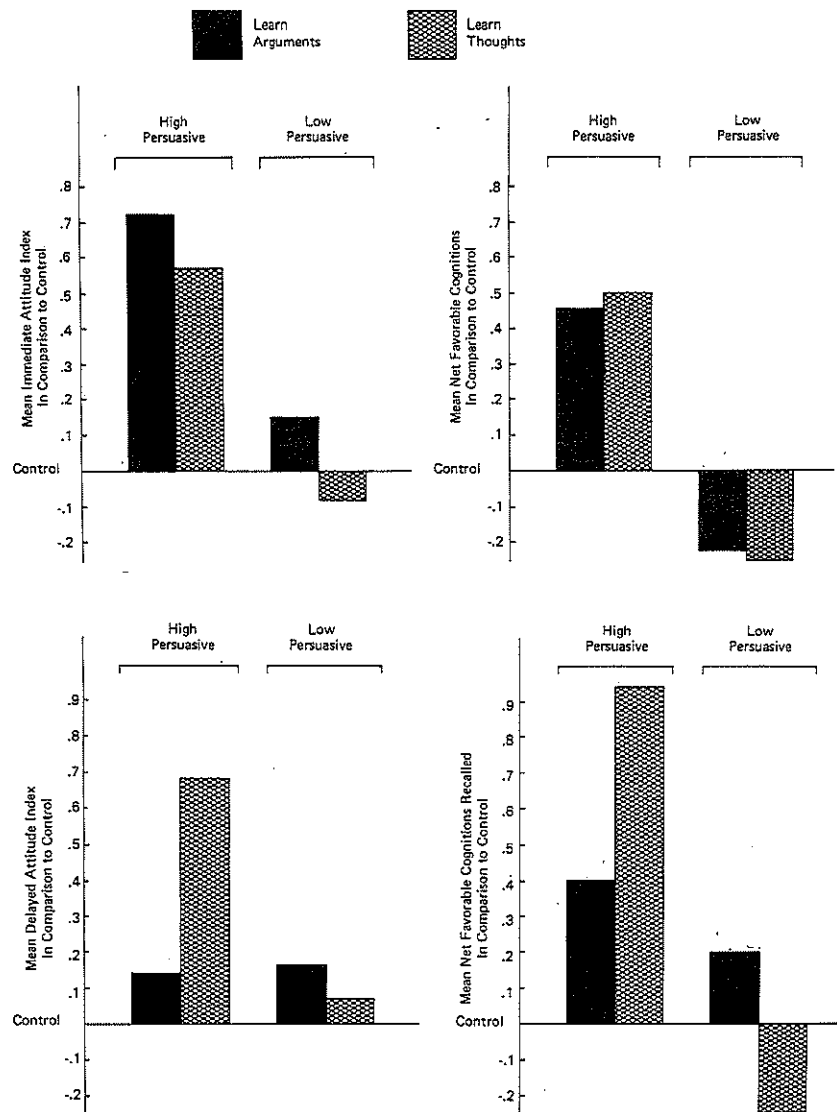


Figure 7-2. Immediate and delayed postmessage attitudes and thoughts as a function of message persuasiveness and learning of arguments and thoughts. *Top panel* shows that subjects were significantly persuaded by the highly persuasive arguments (*left*) and generated more favorable cognitions than controls to these arguments (*right*). Whether subjects learned the arguments or their own thoughts did not affect these initial measures. *Bottom panel* shows that only subjects who memorized their own thoughts to the persuasive arguments persisted in their attitude change (*left*) and recalled more favorable cognitions than controls (*right*). Subjects who memorized the message arguments showed decay of attitudes and favorable cognitions (data from Petty, 1977).

Figure 7.2). In sum, when the initial favorable cognitive responses to the strong message arguments were rehearsed, rendering them more accessible within the attitude schema, the initial attitude change was more enduring than when the specific message arguments were the more accessible component of the attitude schema.

The ELM endorses the cognitive response view that to the extent that the initial message elaborations remain salient, the initial attitude change will persist. However, there are several important differences between the ELM and the cognitive response approach. Perhaps the most crucial concerns the determinants of *initial* persuasion. According to the cognitive response approach, the effects of "most of the traditional independent variables of persuasion . . . (on attitudes) . . . are mediated strictly through their effects on cognitive responses to persuasion" (Greenwald, 1968, p. 168; see also Perloff & Brock, 1980; Petty, Ostrom, & Brock, 1981b). The ELM, of course, postulates *two* routes to persuasion and indicates that attitudes can change in the absence of extended issue-relevant thinking if salient cues are provided in the persuasion context and people lack the requisite motivation and/or ability to engage in message scrutiny. Although the ELM and the cognitive response model share a similar conceptual basis regarding the persistence of persuasion, the ELM does not require the continued salience of the initial message-relevant thoughts for persistence. As we noted in Chapter 1, persistence may also result from a person's perception that a great deal of thinking accompanied attitude formation which induces a reluctance to abandon one's position. Also, relative persistence may result from the repeated pairing of a peripheral cue with a person, object, or issue (e.g., Johnson & Watkins, 1971; Weber, 1972); although these attitudes would still likely be highly susceptible to counterpersuasion because the person would have great difficulty defending his or her position if attacked with strong arguments.

Dissociative Cue Model

In addition to their focus on message learning, Hovland, McGuire, Cook, and others have addressed the operation of augmenting and discounting cues as they relate to the persistence of persuasion. In order to account for the "sleeper effect" (an increase in persuasion over time that may occur when a message is initially paired with a weak source or other discounting cue), Hovland, Lumsdaine, and Sheffield (1949) argued that two independent processes may be involved in persuasion. One process involves *learning* the message arguments (based on message attention and comprehension), but the other involves *accepting* the arguments or the message conclusion (incorporated as "reception" and "yielding" in McGuire's, 1968, model).¹ According to this two-component view, attitude change in response to a persuasive message may be the result of two separate processes:

¹The most recent version of McGuire's information processing model incorporates 12 steps from "exposure" to "post-action" (McGuire, 1985).

learning the message arguments associated with the message conclusion, and associating an acceptance cue with the message conclusion. An acceptance (augmenting) cue (e.g., credible source) is postulated to enhance initial agreement over and above that produced by learning the message arguments alone, whereas a discounting cue (e.g., inexpert source) is thought to initially inhibit the persuasion produced by message learning. It was assumed that over time the link between the cue and the conclusion would decay faster than the link between the message arguments and the conclusion. Because of this, the joint operation of the learning and cue processes could produce a sleeper effect. Specifically, the positive impact of learning message arguments would endure longer than the negative impact of the discounting cue leading to more agreement over time (Cook & Flay, 1978).

The dissociative cue hypothesis and the ELM differ in at least three important respects. The first significant difference has to do with the ELM's focus on issue-relevant *elaboration* rather than message *learning*. The two formulations become somewhat more similar if message "elaboration" is substituted for message "learning" in the dissociative cue formulation. However, at least two essential differences remain.

First, although both models highlight positive and negative cues, the dissociative cue formulation presumes that in a typical persuasion situation, attitudes are affected *jointly* (additively) by salient cues and message processing (learning). In contrast, the ELM highlights the fact that in various persuasion situations (where source and other cues are available prior to or during message presentation) there is a *tradeoff* between message processing and the operation of cues. Thus, attitudes may be affected mostly by cues or by message processing (elaboration). As detailed in Chapter 6, people's attitudes will be affected primarily by various peripheral cues in the persuasion context when motivation and/or ability to process issue-relevant arguments is low. However, people's attitudes will be affected primarily by the quality of the issue-relevant arguments presented when motivation and ability are high. The differential conditions under which either peripheral cues or message processing dominate (e.g., high/low personal relevance, high/low distraction, prior knowledge, need for cognition, etc.) are not considered by the dissociative cue model. The ELM proposes a tradeoff between argument processing and the operation of cues that is moderated by the elaboration likelihood continuum. At the high end of the continuum, argument processing dominates and simple cues have little effect. At the low end of the continuum, cues dominate and little argument processing occurs. In contrast, the dissociative cue model assumes that both cues and argument processing typically operate jointly and additively (Cook, Gruder, Hennigan, & Flay, 1979).

Another important difference between the ELM and the dissociative cue model is that the ELM holds that it is possible for salient contextual factors (including source variables), to affect the manner in which the message

arguments are processed (Petty & Cacioppo, 1981a). In other words, message arguments and certain contextual variables may not have independent effects, but as we have seen previously, various contextual factors may affect the nature of information processing. Specifically, the ELM holds that source and message factors may have *interactive* rather than *additive* effects in some contexts. For example, in Chapter 4 we saw that increasing the number of message sources could increase a person's motivation to process the issue-relevant arguments (see the next chapter for further discussion). In sum, the ELM focuses on message elaboration rather than message learning, postulates a tradeoff between cues and message processing, and suggests that source factors may affect message processing rather than contributing independently to persuasion (i.e., source and message factors may interact).

Although we have just argued that in the typical persuasion situation attitudes are unlikely to be affected in an additive fashion by argument processing and peripheral cues, it is possible for both cues and arguments to affect attitudes *independently* (as specified by the discounting cue model) in certain narrowly defined situations. For example, if the source is identified *after* the message arguments have been processed, then it would be possible for a source cue to affect persuasion over and above the influence of the message. Interestingly, the best evidence for a sleeper effect comes from studies in which a discounting cue is presented *after* rather than prior to message exposure (Gruder, Cook, Hennigan, Flay, Alessi, & Halamaj, 1978; Hovland & Weiss, 1951; Pratkanis & Greenwald, 1985). However, according to the ELM, a sleeper effect should still be very difficult to produce even if a negative cue follows a message. This is because if the message is on a topic of low personal relevance or knowledge, for example, the postmessage cue may be effective, but the message will have induced insufficient processing to lead to persistence. If the message is on a topic of high personal relevance or knowledge, however, although the message may induce considerable processing, many negative cues (e.g., an unattractive source) presented after the message may be judged irrelevant and therefore ignored. Although the sleeper effect is of considerable conceptual significance, obtaining it may require conditions that are infrequently present in either the "real world" or in persuasion research.

Three Process Model

Kelman (1958, 1961) has developed a theory of attitude persistence that is based on specifying the antecedents and consequents of three different attitude change processes. In Kelman's typology, attitudinal *compliance* occurs when attitudes change as a result of a person wanting to obtain rewards or avoid punishments under the control of some powerful source. Change is postulated to persist only to the extent to which the powerful source can monitor the person's attitudes. Attitudinal *identification* occurs when attitudes change as a result of a person wanting a satisfying

relationship with or wanting to view him or herself as similar to some attractive source. Change is thought to persist only to the extent to which the person's relationship with the attractive source remains a salient goal. Attitudinal *internalization* occurs when attitudes change as a result of a person accepting new information that is perceived to be inherently conducive to the maximization of preexisting values. According to Kelman (1961), a crucial dimension for acceptance of such information, and thus for internalization, "is the agent's credibility, that is his relation to the (message) content" (p. 65). Since attitude change through internalization is independent of the power of, or one's personal relationship with a source, it is believed to persist. Although Kelman (1958) has conducted one study that is consistent with the three process theory (see discussion in Chapter 8), other research has not been as supportive. In particular, there is little evidence for the view that source credibility generally enhances persistence (Cook & Flay, 1978).

The ELM shares the view that attitude changes based on "internalization" of issue-relevant information are more likely to persist than those based on certain cues (e.g., source attractiveness), but the formulations differ in the antecedents of these different processes. Kelman ties the type of persuasion mostly to the nature of the message source. Thus, expert sources produce internalization, attractive sources produce identification, and powerful sources produce compliance. According to the ELM, expertise, attractiveness, and power may all serve as simple cues under some circumstances, producing short-lived agreement resulting from little issue-relevant thinking. Under other circumstances, however, issue-relevant thinking may induce relatively enduring persuasion regardless of source expertise, attractiveness, or power. According to the ELM, the crucial determinant of persistence is not the source of the message, but the person's general motivation and ability to process issue-relevant arguments (see Figure 1-1). Also, although Kelman's internalization is similar to what we have called persuasion via the "central" route, and "identification" and "compliance" would qualify as peripheral approaches, the ELM does not tie the effectiveness of sources to these particular processes. For example, in some situations the use of an attractive source may induce persuasion via the process of identification as Kelman suggests. In others, an attractive source may induce change by invoking the balance principle (Insko, 1981), or may influence agreement by triggering an attractiveness heuristic (Chaiken, 1986). Although it is important to specify why attractiveness serves as a cue in different situations (e.g., Mills & Harvey, 1972), the ELM views all of these processes as peripheral in that they are postulated to operate under similar conditions (i.e., low elaboration likelihood) and have similar consequences (e.g., relative lack of persistence compared with changes based on issue-relevant argumentation).

Attitude-Behavior Link

Enhanced Thinking Produces Attitude-Behavior Consistency

In the previous section we provided support for the view that attitude changes based primarily on thoughtful consideration (or self-generation) of issue-relevant arguments produced more enduring persuasion than changes based primarily on simple cues in the persuasion context. Research is also consistent with the view that attitudes formed or changed via the central route are more predictive of behavior than attitudes based on peripheral cues.

As we noted previously, one effortful form of issue-relevant processing occurs when attitude change results from the self-generation of arguments, as in research on role playing (e.g., Watts, 1967). These changes, then, should be especially predictive of behavior. In a relevant program of research, Fazio and Zanna (1981) have explored the consequences of attitudes formed via direct rather than indirect experience. When an attitude is formed through direct personal experience, the attitude is necessarily based on self-generated information. When an attitude is based on indirect experience (i.e., a message from others), less effortful processing typically may be involved. In some sense then, the distinction between direct and indirect experience is analogous to the distinction between attitudes based on role playing (i.e., self-generation of arguments) versus passive exposure.

In a prototypical study on direct versus indirect experience with the attitude object, Regan and Fazio (1977, Experiment 2) varied the manner in which subjects' attitudes toward five intellectual puzzles were formed. Half of the subjects were presented with previously solved examples of each puzzle by an experimenter who explained the directions and solutions (indirect experience). A second group of subjects was given the opportunity to work the puzzles themselves. These subjects had direct behavioral experience, but presumably also had to self-generate the positive or negative attributes of the puzzles because the experimenter did not supply them. Subsequently, all subjects rated how much they liked the puzzles, and then were given 15 min. to work any puzzles they desired. The attitudes of subjects in the direct experience (self-generation) group proved better predictors of behavior toward the puzzles than the attitudes of subjects with only passive exposure to information. For example, attitudes of the direct experience subjects predicted the order in which the puzzles were attempted ($r = 0.51$) better than the attitudes of subjects with indirect experience ($r = 0.22$). Other research employing different attitude objects and behaviors has also supported the view that attitudes formed via direct experience with the attitude object predict behavior better than attitudes based on indirect information (see review by Fazio, 1985). The ELM suggests that one reason

for this is that attitude formation based on direct experience may typically require greater elaboration of the merits of the object (e.g., puzzle) than attitude formation based on passive exposure.

Consider another study by Sivacek and Crano (1982), which explored the attitude-behavior relationship in groups of people who either did or did not have a personal stake in an issue. The attitudes of interest in their study concerned an impending statewide referendum to raise the legal drinking age from 19 to 21. One group of subjects studied would be personally affected by the proposal (i.e., those who would not be 21 by the time the proposed law went into effect), and one group would not be affected personally. Even though both groups of subjects expressed equally strong attitudes against the proposal, more people who would be personally affected by the proposal agreed to engage in behaviors consistent with their negative attitudes than people who would not be personally affected. If we can assume that the attitudes of the high relevance group were more likely formed via the central route (i.e., extensive issue-relevant thought), whereas the attitudes of those who would not be affected were more likely formed via the peripheral route (e.g., identifying with the opinions of their friends), then the results of this study are consistent with the view that attitudes formed via the central route are more predictive of behavior than attitudes formed via the peripheral route.

Finally, consider a series of studies by Davidson, Yantis, Norwood, and Montano (1985) that examined the attitude-behavior relation in groups of people who differed in the amount of information they possessed about the attitude object. In one study, subjects' voting intentions toward the two candidates for mayor in a local election were assessed 4 to 14 days prior to the election. In addition, the subjects were asked to list the information and beliefs that they possessed about the candidates. Then, following the election, subjects were telephoned and asked whether they voted in the election, and if so, for whom. Analyses were confined to people who said they voted. The major result of this study was that the amount of information a person had about the candidates moderated the relationship between preelection intentions and reported voting in the election. Specifically, regression analyses indicated that as the amount of information about the candidates increased, so did the correlation between intentions and behavior. These results were replicated in two additional studies in which the amount of information people had about an attitude object was assessed in a self-report format (7-point scale ranging from "completely uninformed" to "completely informed") rather than with a thought-listing. Furthermore, in these studies, the amount of information remained a moderator of intention-behavior consistency even after controlling for previous experience with the attitude object and confidence in one's intentions. If we make the reasonable assumption that people with more information about an attitude object were more likely to have formed their attitudes and

intentions via the central rather than the peripheral route, these results are consistent with the ELM.

Testing the ELM Analysis of Attitude-Behavior Consistency

Enhanced Elaboration Increases Attitude-Behavior Consistency

In general, the accumulated attitude-behavior research is consistent with the view that the more cognitive effort that is involved in attitude formation, the greater the utility of the attitude in predicting behavior (Cialdini, Petty, & Cacioppo, 1981). In our own research, we have also found that conditions fostering a high elaboration likelihood produce higher attitude-behavior correlations than conditions in which the elaboration likelihood is low. In one study described in Chapter 6, for example, we exposed subjects to mock magazine advertisements for a disposable razor under conditions of either high or low personal relevance, (Petty, Cacioppo, & Schumann, 1983). Recall that under high relevance, attitudes were affected primarily by the issue-relevant arguments presented for the product, but under low relevance attitudes were affected more by peripheral source cues (see Figure 6-3). Importantly, in addition to assessing product attitudes in this study, we also asked subjects to rate how likely they were to purchase the product (behavioral intentions). Overall, the attitude-intention correlation under high relevance was 0.59, whereas under low relevance it was 0.36 ($p < 0.07$).

Just as increasing motivation to process issue-relevant arguments should enhance the utility of attitudes in predicting behavior, so too should enhancing ability to process the message. In one relevant study, we exposed subjects to advertisements containing strong arguments for a new pen either 1, 4, or 8 times in the context of a simulated television program (Schumann, Petty, & Cacioppo, 1986). Each repetition of the message gives subjects an additional opportunity to consider the product-relevant information (see Chapter 3). After message exposure, subjects rated their attitudes toward the advertised pen, their likelihood of purchasing this brand in the near future, and the amount of time they spent thinking about the product during the program. Subjects reported engaging in more thought about the product as repetition increased, and the attitude-intention correlation also improved significantly with repetition. Specifically, attitudes and intentions toward the pen product correlated 0.65 with one exposure. By four exposures, the correlation increased to 0.78 and remained at this high level at eight exposures (0.75).

Need for Cognition and Attitude-Behavior Consistency

The ELM holds that situational variables can moderate the attitude-behavior relationship by affecting the likelihood that attitudes are formed or changed via the central or peripheral routes. Relevant dispositional

variables should operate in a similar manner. For example, we have provided considerable evidence for the view that the need for cognition scale (NCS; Cacioppo & Petty, 1982) allows at least gross distinctions to be made between individuals who differ chronically in their tendency to engage in and enjoy effortful cognitive endeavors (see Chapter 2). Furthermore, the cumulative research on NC and persuasion indicates that as expected, the attitudes of high need for cognition (HNC) individuals are more likely to be affected by issue-relevant thinking (central route; see Chapter 4), whereas the attitudes of low need for cognition (LNC) individuals are more likely to be affected by peripheral cues (see Chapter 6). Because of this, it would be expected that the attitudes of HNC individuals would be more predictive of their behavior than the attitudes of LNC persons. In order to test this hypothesis, we conducted a study during the 1984 presidential campaign (Cacioppo, Petty, Kao, & Rodriguez, in press, Experiment 2).

Method. A survey was administered to a large number of undergraduates 8 weeks prior to the 1984 presidential election. The survey included measures of attitudes toward the presidential tickets (Mondale/Ferraro; Reagan/Bush), reported thought and knowledge about the candidates, voting intentions, and level of need for cognition. A subset of individuals differing widely in their NC scores (falling in the top and bottom thirds of the distribution) were selected for further study. Of these subjects, 61 constituted the "knowledge subsample," and 108 constituted the "attitude-behavior subsample."

Subjects in the knowledge subsample group were invited to the laboratory either 3 to 5 weeks prior to the election ($N = 33$) or 3 to 5 weeks after the election ($N = 28$). During the session, subjects were asked to list everything they knew about the presidential candidates. Subjects were given 5 min. to list one fact per line and were told that spelling and grammar were unimportant. The number of unique items of information listed by the subject served as the primary dependent measure of knowledge.

Subjects in the attitude-behavior subsample were contacted by telephone from 1 to 3 days following the election. Subjects were asked if they had voted, and if so, for whom. Subjects who indicated that they had not voted were asked to indicate for whom they would have voted. Of the 56 LNC individuals who were contacted, 41 (73%) reported having voted, and 43 (83%) of the 52 HNC individuals reported having voted (*n.s.*).

Results. A 2 (Need for cognition: high or low) $\times$ 2 (Time of measurement: before or after election) between-subjects ANOVA on the amount of information possessed by subjects in the knowledge subsample revealed only a main effect for NC. As expected, HNC individuals generated more facts about the presidential candidates ($M = 8.13$) than LNC persons ($M = 6.10$).

Similar mixed model ANOVAS on the data from subjects in the attitude-behavior subsample revealed main effects for both Time and NC. Subjects reported having thought more about the candidates, having more knowledge about, and more confidence in their selections after than before the election. Also, HNC individuals reported having thought more about the candidates and knowing more about them than LNC persons. One interaction revealed that HNC individuals reported more thinking about the candidates before the election than LNC persons, but after the election both groups reported similar amounts of thinking. Finally, HNC individuals reported more favorable attitudes toward the Mondale/Ferraro ticket than LNC individuals.

To test our major hypothesis that the preelection attitudes of HNC, in contrast to LNC, subjects would be more predictive of subsequent reports of voting, the correlation between a relative preference index (difference score between attitudes toward the two tickets; positive numbers favored Mondale/Ferraro) and reported voting (1 = Reagan/Bush; 2 = Mondale/Ferraro) was calculated separately for the HNC and the LNC groups. The preelection preference index predicted postelection reports better for HNC ($r = 0.87$) than LNC ($r = 0.46$) individuals. Because these correlations included all subjects, whether they reported actually voting or not, the correlations were recomputed for the subsample that reported actually voting. Again, attitudes predicted behavior better for HNC ($r = 0.86$) than LNC ($r = 0.41$) individuals.

Conclusions

Before proceeding to the next consequence of elaboration, it is noteworthy that our argument that the more issue-relevant elaboration involved in attitude formation or change, the greater the attitude-behavior correlation, appears to conflict with a recent claim by Wilson, Dunn, Bybee, Hyman, and Rotondo (1984) that analyzing reasons for one's attitudes *reduces* attitude-behavior consistency. In the research supporting the contention of Wilson et al., however, one effect of having subjects think about the reasons behind their attitudes was to produce a change in attitudes. Thus, Wilson and his colleagues have compared the ability of an *initial* attitude to predict behavior with the ability of a *changed* attitude. Their results indicated that the new attitude was less predictive than the old one. Importantly, the ELM addresses a comparison between two initial attitudes (one formed via the central route and one formed via the peripheral route) or two newly changed attitudes (one changed via the central and one changed via the peripheral route). The ELM predicts that the central attitudes will predict behavior better than comparable attitudes formed or changed via the peripheral route.² As we have seen, the available evidence is consistent with the view

²Wilson (1986) argues that analyzing reasons may also reduce attitude-behavior consistency even if attitudes are not changed, however. He further argues that this effect may occur primarily when (a) the attitudes in question are affectively based

that when the experimental conditions or dispositional factors enhance people's motivation or ability to elaborate issue-relevant information, the attitudes formed or changed predict intentions and behaviors better than when the elaboration likelihood is low (see also, Kallgren & Wood, 1985; Pallak, Murrone, & Koch, 1983; Sandelands & Larson, 1985).

Resistance to Counterpersuasion

Enhanced Thinking Produces Resistance

A third consequence of the route to persuasion postulated by the ELM is that attitudes formed or changed via the central route should be more resistant to counterpropaganda than attitudes formed or changed via the peripheral route. Importantly, the *resistance* of an attitude to attack is conceptually distinct from the temporal *persistence* of an attitude. Thus, some attitudes may be highly persistent, but only if they are not challenged. Other attitudes may be very transient even in a vacuum. Likewise, it is possible for some attitudes to be very resistant to change, but only in the short term. Despite the conceptual independence of persistence and resistance, we have already outlined the reasons why the ELM holds that usually these two qualities will go together. Attitudes based on extensive issue-relevant thinking will tend to be both persistent and resistant, whereas attitudes based on peripheral cues will tend to be transient and susceptible to counterpersuasion.

Attitudes for which persistence and resistance do *not* go together provide an intriguing target of study. Perhaps the most dramatic example of the possible independence of persistence and resistance is found in cultural truisms. Truisms such as "you should brush your teeth after every meal" tend to be highly persistent in a vacuum, but very susceptible to influence if attacked. As McGuire (1964) noted, people have very little practice in defending these beliefs because they have never been attacked. Furthermore, the ELM would contend that these beliefs are highly susceptible to persuasion because they were probably formed with very little issue-relevant thinking. It is likely that people come to accept unquestioningly the various cultural truisms sometime during childhood. The truisms are presented many times during development by powerful, likable, and expert sources (e.g., parents, teachers, television characters) with little or no justification. This continual pairing of a belief with a positive source cue may result in a relatively persistent attitude, but one that cannot be defended when subsequently attacked.

(e.g., attitude toward a dating partner) rather than cognitively based (e.g., attitude toward a presidential candidate), and (b) the cognitive analyses are conducted on a short-term rather than a long-term basis. These very interesting suggestions warrant further investigation.

Most research on attitudinal resistance has focused on how various "treatments" can help bolster an attitude that a person already has. In an important program of research, McGuire (1964) has provided impressive evidence for the view that attitudes toward truisms can be made more resistant by providing people with the requisite motivation and/or ability to counterargue opposing messages. For example, in an early study on the "inoculation effect," McGuire and Papageorgis (1961) provided subjects with a "supportive defense" (which presented arguments supporting subjects' original beliefs) on one cultural truism, a "refutational defense" (which presented and refuted arguments *against* subjects' beliefs) on another truism, and no defense on a third truism. Two days later, subjects were exposed to messages that attacked their initial beliefs on all three truisms. The results showed that both defenses were helpful in inducing resistance to the counterpropaganda, but the refutational defense (inoculation treatment) was significantly more effective. The underlying logic of McGuire's inoculation theory is that a threat to a previously unchallenged belief, provided in the refutational defense, motivates the person to defend that belief when it is assailed in the future. People have probably never heard certain truisms impugned, and may not be able to conceive of them as being susceptible to attack. A refutational defense disabuses people of this notion, motivating them to bolster the assailed belief and giving them practice in defending it.

Research employing cultural truism topics has replicated McGuire's finding that refutational defenses are superior to supportive defenses in conferring resistance (e.g., Suedfeld & Borrie, 1978). However, research not employing cultural truisms has tended to find that refutational and supportive defenses confer similar degrees of resistance (e.g., Adams & Beatty, 1977; Pryor & Steinfatt, 1978). When the issue employed is not a cultural truism, and people are therefore not operating under the assumption that their belief is invulnerable, the refutational defense should not have any unique motivating power, and thus would not necessarily be more effective than a supportive defense.

Testing the ELM Resistance Predictions

Importantly, the work on resistance using McGuire's inoculation paradigm and others (e.g., Burgoon, Cohen, Miller, & Montgomery, 1978; see review by Smith, 1982) has focused on how an initial belief held by a person can be made more resistant by providing some treatment that enhances the person's motivation and/or ability to counterargue a subsequent opposing communication. This work is consistent with the ELM in that it demonstrates that attitudes can be made more resistant by motivating or enabling people to engage in additional thought about the reasons or arguments supporting their attitudes. However, the ELM makes unique predictions about the differential resistance of attitudes formed via different routes.

Specifically, the ELM postulates that people who come to accept an initial position because of a peripheral cue (e.g., source expertise) should be generally more susceptible to an attacking message than people who adopt the same initial position based on a careful scrutiny and elaboration of the message arguments. We tested this prediction in a recent study (Petty, Cacioppo, Haugtvedt, & Heesacker, 1986; Experiment 2).

Method. A total of 113 students participated in the study. The experimental conditions comprised a 2 (Extent of elaboration: low or high) $\times$ 2 (Message direction: pro or anti) mixed design with extent of processing serving as a between-subjects variable and message direction serving as a within-subjects factor. Several control groups were also included (see below). Undergraduate students were recruited to participate in a "media evaluation" study. Upon arriving at the lab they were told that the psychology department was helping the Journalism School obtain ratings of audiotapes submitted by applicants to the broadcasting program. They were further told that in order to obtain ratings simulating those that would be obtained under normal radio listening conditions in which distractions were present, they would be asked to monitor the quadrant in which an X appeared on a screen before them. Subjects expected to listen to and rate five different announcers during the session.

All experimental subjects initially listened to an irrelevant message and then to a message attributed to three college professors from prestigious universities that presented seven strong arguments for the imposition of comprehensive exams as a requirement for college graduation (pro-exam communication). Following this, subjects were exposed to another irrelevant message, and then to a message containing seven arguments against the imposition of senior comprehensive exams that was attributed to another prestigious college professor (anti-exam communication).

Half of the experimental subjects were randomly assigned to hear the first key (pro-exam) message under conditions that would inhibit message elaboration. These subjects were told, via an introduction on the tape, that the announcer for the message was from a distant university and that the proposal to institute the exams was proposed for this distant school (i.e., low personal relevance conditions that were designed to reduce motivation to process the message; Petty & Cacioppo, 1979b; see Chapter 4). In addition, for these subjects the Xs flashed on the screen once every 5 s (a high distraction task that was designed to disrupt processing; Petty, Wells, & Brock, 1976; see Chapter 3). The remaining experimental subjects were assigned to hear the pro-exam message under conditions that were designed to be more facilitative of argument processing. For these subjects, the announcer for the message was from their own university and they were led to believe that the proposal would affect students at their own school (high personal relevance). Furthermore, the Xs flashed on the screen once every 15 s (low distraction). For all subjects, the anti-exam message was attributed

to an announcer from a distant school, and Xs flashed during this message at the low distraction rate. Based on the previous research reported in this volume and pilot testing for this study, we expected subjects in the low elaboration conditions to be influenced primarily by the positive source cues associated with the issue; but subjects in the high elaboration conditions were expected to be influenced primarily because of their processing of the stronger issue-relevant arguments.

Following each message, experimental subjects rated the student announcer, the source of the information contained in the communication (i.e., the college professors), and also provided their own opinions about each of the issues "since your own opinion on the issue discussed may influence your ratings of the announcer." Thus, subjects gave their opinions about senior comprehensive exams twice—once following the pro-exam message and once following the anti-exam message. After the final communication was presented, all subjects were given a brief period of time to list the thoughts they recalled having to the pro-exam message and then the anti-exam message. A no message control group was also included in the study to assess the effectiveness of the pro-exam message conditions in producing absolute attitude change.³

Results. Subjects in both the high and the low elaboration experimental groups were significantly (and equally) more favorable toward senior comprehensive exams after exposure to the pro-exam message than were control subjects who did not receive the pro-message. Importantly, correlational analyses supported the view that even though the amount of change was similar in both groups, the processes inducing change were different and as expected. Specifically, the attitudes of high elaboration subjects were correlated with the number of favorable issue-relevant thoughts that they generated about senior comprehensive exams ($r = .38, p < .10$) but were unrelated to their ratings of the source of the information ($r = .22$). In contrast, the attitudes of low elaboration subjects were uncorrelated with the number of favorable thoughts generated ($r = .11$), but were predicted by their ratings of the source of the information about the exams ($r = .59, p < .01$).

The major attitude results from the study are depicted in Figure 7-3. A 2 (Extent of elaboration) $\times$ 2 (Message direction) ANOVA on attitudes revealed two significant effects. First, a main effect for message indicated

³Additional groups of control subjects heard the initial pro-exam message under high or low elaboration conditions, provided their opinions, and then responded to the senior comprehensive exam issue again *without* hearing the anti-exam message. The results from these controls (graphed in the bottom panel of Figure 7-3) revealed only a main effect for time indicating that both groups of subjects showed a very small decay on the second attitude assessment.

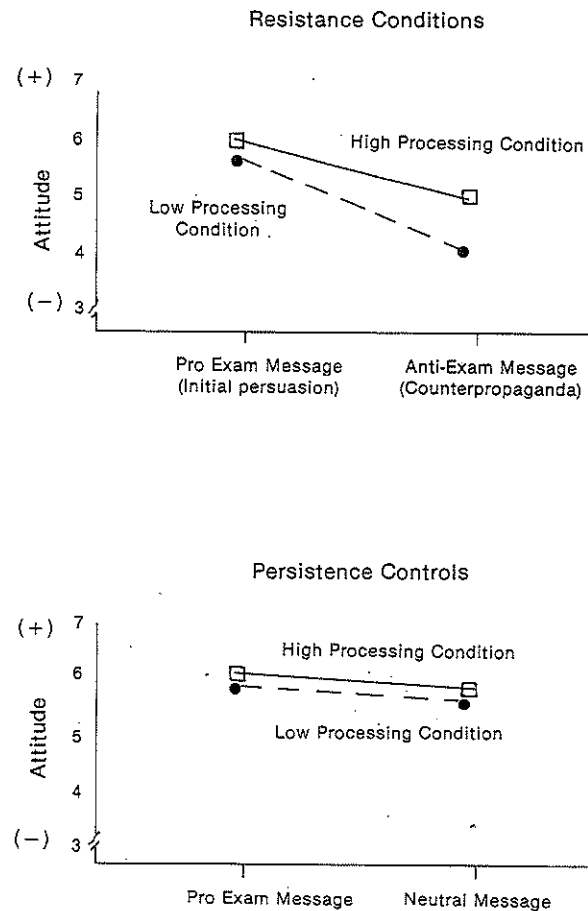


Figure 7-3. Attitudes after an initial message and subsequent counterpropaganda as a function of the initial level of message processing. *Top panel* shows that attitude changes induced under high processing conditions (high relevance, low distraction) showed greater resistance to counterpropaganda than comparable attitude changes induced under low processing conditions (low relevance, high distraction). *Bottom panel* depicts control conditions showing that in the absence of counterpropaganda, the attitude changes induced under high and low processing conditions showed a similar small amount of decay (data from Petty, Cacioppo, Haugtvedt, & Heesacker, 1986, Experiment 2).

that subjects became less favorable toward the exams after exposure to the anti-exam message. However, an Extent of elaboration $\times$ Message direction interaction provided support for our major hypothesis: subjects whose attitudes were changed under high elaboration conditions were more resistant to the counterpersuasive message than subjects who initially changed the same amount, but under low elaboration conditions. In short, subjects who became more favorable toward senior comprehensive exams because they elaborated the issue-relevant arguments maintained their favorable positions in the face of counterpropaganda better than did subjects who formed quantitatively similar attitudes on the basis of the positive source cues.

Conclusions

Although there has been considerable interest in McGuire's (1964) program of research on different types of treatments that increase the likelihood that an already existing attitude will resist attack, little attention has been paid to the factors that determine whether a newly formed attitude will or will not prove resistant to subsequent propaganda. The ELM clearly indicates that attitude changes that take place when situational (e.g., high relevance) or individual difference (e.g., high need for cognition) factors foster issue-relevant elaboration should prove more resistant in the face of counterpropaganda than comparable attitude changes that result from simple cues under low elaboration conditions. Interestingly, and consistent with the ELM, recent research has suggested that high knowledge subjects who form an initial judgment about some material may even be more resistant to conflicting messages that they generate themselves than are people who have low knowledge on the issue (Hirt & Sherman, 1985).

Retrospective

In this chapter we discussed three major consequences of the route to persuasion. First, evidence was reviewed that was consistent with the view that the more issue-relevant thinking accompanying attitude formation or change, the more likely the attitude was to persist over time. This prediction and others from the ELM was contrasted with three alternative models of attitude persistence. Second, research was reviewed which indicated that attitudes based on issue-relevant thinking or extensive issue-relevant information were more predictive of behavior than attitudes based on peripheral cues. Finally, we noted that the available evidence supported the view that issue-relevant thinking induces resistance to subsequent counterattitudinal propaganda. Now that we have addressed the major antecedents and consequences of the two routes to persuasion, in the next chapter we examine some complicating factors in the ELM.